Exercise C - II

SUBJECTIVE / INTEGER TYPE QUESTIONS

1. Let $f(x) = 1 - x + x^2 - x^3 + ... + x^{16} - x^{17} = a_0 + a_1 (1 + x) + a_2 (1 + x)^2 + ... + a_{17} (1 + x)^{17}$,

find the value of a_2.

माना $f(x) = 1 - x + x^2 - x^3 + ... + x^{16} - x^{17} = a_0 + a_1 (1 + x) + a_2 (1 + x)^2 + ... + a_{17} (1 + x)^{17}$ तो a_2 का मान ज्ञात कीजिए।

Ans. 816

Sol. $f(x) = 1 - x + x^2 - x^3 + ... + x^{16} - x^{17}$

$$= \frac{1\left(1-(-x)^{18}\right)}{1-(-x)}$$

$$= \frac{1-x^{18}}{1+x}$$

$$\frac{1-x^{18}}{1+x} = a_0 + a_1(1+x) + a_2(1+x)^2 + ... + a_{17}(1+x)^{17}$$

$1 - x^{18} = a_0 (1 + x) + a_1 (1 + x)^2 + a_2 (1 + x)^3 + ... + a_{17} (1 + x)^{18}$

Let $1 + x = t$

$x = t - 1$

$1 - (t - 1)^{18} = a_0 t + a_1 t^2 + a_2 t^3 + ... + a_{17} t^{18}$

So, we need to find coefficient of t^3

$1 - (1 - t)^{18}$

$1 - \{{}^{18}C_0 - {}^{18}C_1 t_1 + {}^{18}C_2 t^2 - {}^{18}C_3 t^3 + ...\}$

So ${}^{18}C_3 \Rightarrow \frac{18}{3} \times \frac{17}{2} \times \frac{16}{1} = 816$

2. Let $a = \left(4^{1/401} - 1\right)$ and let $b_n = {}^nC_1 + {}^nC_2 \cdot a + {}^nC_3 \cdot a^2 + ... + {}^nC_n \cdot a^{n-1}$.

Find the value of $(b_{2006} - b_{2005})$.

माना $a = \left(4^{1/401} - 1\right)$ और माना $b_n = {}^nC_1 + {}^nC_2 \cdot a + {}^nC_3 \cdot a^2 + ... + {}^nC_n \cdot a^{n-1}$ है।

तो $(b_{2006} - b_{2005})$ का मान ज्ञात कीजिए।

Ans. 2^{10}

Sol. $a = 4^{\frac{1}{401}} - 1$

$b_n = {}^nC_1 + {}^nC_2\, a + {}^nC_3\, a^2 + ... + {}^nC_n\, a^{n-1}$

$ab_n = {}^nC_1\, a^1 + {}^nC_2\, a^2 + {}^nC_3\, a^3 + ... + {}^nC_n\, a_n$

$+ \qquad +$

${}^nC_0 \qquad {}^nC_0$

$ab_n + {}^nC_0 = [1 + a]^n$

$ab_n = (1 + a)^n - 1$

$n = 2006$

$$ab_{2006} = \frac{(1+a)^{2006}}{a} - \frac{1}{a}$$

$$b_{2005} = \frac{(1+a)^{2005}}{a} - \frac{1}{a}$$

$$b_{2006} - b_{2005} = \frac{(1+a)^{2006}}{a} - \frac{1}{a} - \frac{(1+a)^{2005}}{a} + \frac{1}{a}$$

$$= \frac{(1+a)^{2006} - (1+a)^{2005}}{a}$$

$$= \frac{[1+a]^{2005}\{1+a-1\}}{a}$$

$\Rightarrow (1 + a)^{2005}$

$1 + a = 4^{\frac{1}{401}}$

$(1+a)^{2005} = 4^{\frac{2005}{401}} = 4^5 = 2^{10}$

3. Let $a = 3^{\frac{1}{223}} + 1$ and let $f(n) = {}^nC_0 \cdot a^{n-1} - {}^nC_1 \cdot a^{n-2} + {}^nC_2 \cdot a^{n-3} - ..... + (-1)^{n-1} \cdot {}^nC_{n-1} \cdot a^0$.

If the value of $f(2007) + f(2008) = 9k$ where $k \in N$, then find k.

माना $a = 3^{\frac{1}{223}} + 1$ और माना $f(n) = {}^nC_0 \cdot a^{n-1} - {}^nC_1 \cdot a^{n-2} + {}^nC_2 \cdot a^{n-3} - \ldots\ldots + (-1)^{n-1} \cdot {}^nC_{n-1} \cdot a^0$.

यदि $f(2007) + f(2008) = 9k$ जहां $k \in N$ तो k का मान ज्ञात कीजिए।

Ans. 2187

Sol. $a = 3^{\frac{1}{223}} + 1$

$f(n) = {}^nC_0\, a^{n-1} - {}^nC_1\, a^{n-2} + {}^nC_2\, a^{n-3} - \ldots\ldots + (-1)^{n-1}\, {}^nC_{n-1}\, a_0$

$af(n) = {}^nC_0\, a^n - {}^nC_1\, a^{n-1} + {}^nC_2\, a^{n-2} - {}^nC_n\, a^0 \ldots\ldots + (-1)^{n-1}\, {}^nC_{n-1}\, a + (-1)^n\, {}^nC_n\, a^0$

$+$

$(-1)^n \cdot {}^nC_n\, a^0$

$= [a-1]^n$

$af(n) + (-1)^n = (a-1)^n$

$af(2007) - 1 = (a-1)^{2007}$

$af(2008) + 1 = (a-1)^{2008}$

$a\{f(2007) + f(2008)\} = (a-1)^{2007}\{1 + (a-1)\}$

$f(2007) + f(2008) = (a-1)^{2007}$

$\Rightarrow 3^{\frac{2007}{223}} \Rightarrow 3^9 = 9k$, $k = 2187$

4. Let $P = \left(2+\sqrt{3}\right)^5$ and $f = P - [P]$, where $[P]$ denotes the greatest integer function.

Find the value of $\left(\frac{f^2}{1-f}\right)$.

माना $P = \left(2+\sqrt{3}\right)^5$ तथा $f = P - [P]$, जहां $[P]$ महत्तम पूर्णांक फलन है। तो $\left(\frac{f^2}{1-f}\right)$ का मान ज्ञात कीजिए।

Ans. 722

Sol. $P = \left(2+\sqrt{3}\right)^5$

$f = P - [P]$

$\frac{f^2}{1-f}$

$I + f = \left(2+\sqrt{3}\right)^5 = {}^5C_0 2^5 + {}^5C_1 2^4\sqrt{3}^1 + {}^5C_2 2^3\sqrt{3}^2 + ... + {}^5C_5\left(\sqrt{3}\right)^5$

$F = \left(2-\sqrt{3}\right)^5 = {}^5C_0 2^5 - {}^5C_1 2^4\sqrt{3} + ... - {}^5C_5\left(\sqrt{3}\right)^5$

$I + f + F = 2[{}^5C_0\, 2^5 + {}^5C_2\, 2^3\, 3 + {}^5C_4\, 2\cdot 3^2]$

$= 2\,[32 + 10\cdot 24 + 5\cdot 18]$

$= 724$

$0 < f < 1$

$0 < F < 1$

$0 < f + F < 2$

$\Rightarrow f + F = 1$

$I = 723$

$F = 1 - f$

$([P] + f)\, F = 1$

$$[P] = \frac{1}{F} - f$$

$$= \frac{1}{1-f} - f$$

$$= \frac{1-f+f^2}{1-f}$$

$$723 = 1 + \frac{f^2}{1-f}$$

$$\frac{f^2}{1-f} = 722$$

5. Prove that :
सिद्ध कीजिये

(a) $\frac{C_1}{C_0} + \frac{2C_2}{C_1} + \frac{3C_3}{C_2} + ... + \frac{n.C_n}{C_{n-1}} = \frac{n(n+1)}{2}$

(b) $C_0 + \frac{C_1}{2} + \frac{C_2}{3} + ... + \frac{C_n}{n+1} = \frac{2^{n+1}-1}{n+1}$

(c) $2 \cdot C_o + \frac{2^2 \cdot C_1}{2} + \frac{2^3 \cdot C_2}{3} + \frac{2^4 \cdot C_3}{4} + ... \frac{2^{n+1} \cdot C_n}{n+1} = \frac{3^{n+1}-1}{n+1}$

(d) $C_o - \frac{C_1}{2} + \frac{C_2}{3} - ..... + (-1)^n \frac{C_n}{n+1} = \frac{1}{n+1}$

Ans.

Sol. (a) $\frac{C_1}{C_0} + \frac{2C_2}{C_1} + ... + \frac{nC_n}{C_{n-1}}$

$$\sum_{r=1}^{n} \frac{r \, {}^nC_r}{{}^nC_{r-1}}$$

$$\sum_{r=1}^{n} n \frac{{}^{n-1}C_{r-1}}{{}^nC_{r-1}}$$

$$n\sum_{r=1}^{n} \frac{\frac{(n-1)!}{(n-1-r+1)!(r-1)!}}{\frac{n!}{(n-r+1)!(r-1)!}}$$

$$n\sum \frac{(n-1)!}{n!} \times \frac{(n-r+1)!}{(n-r)!}$$

$$n\sum \frac{1}{n} \times (n-r+1)$$

$$\sum_{r=1}^{n} (n-r+1)$$

$$n^2 - \frac{n(n+1)}{2} + n$$

$$\frac{2n^2 - n^2 - n + 2n}{2}$$

$$\frac{n^2+n}{2} \Rightarrow \frac{n(n+1)}{2}$$

(b) ${}^nC_0 + \frac{{}^nC_1}{2} + \frac{{}^nC_2}{3} + ... + \frac{{}^nC_n}{n+1}$

$$\sum_{r=0}^{n} \frac{{}^{n}C_r}{r+1}$$

$$\frac{1}{n+1}\sum_{r=0}^{n} \frac{n+1}{r+1}.{}^{n}C_r$$

$$\frac{1}{n+1}\sum_{r=0}^{n} {}^{n+1}C_{r+1}$$

$$\frac{1}{n+1}\left\{{}^{n+1}C_1 + {}^{n+1}C_2 + ... + {}^{n+1}C_{n+1}\right\}$$

$$+ \qquad\qquad - {}^{n+1}C_0$$

$${}^{n+1}C_0$$

$$\frac{1}{n+1}.\left\{2^{n+1}-1\right\}$$

(c) $$2.{}^{n}C_0 + 2^2.\frac{{}^{n}C_1}{2} + 2^3.\frac{{}^{n}C_2}{3} + ... + \frac{2^{n+1}\ {}^{n}C_n}{n+1}$$

$$\sum_{r=0}^{n} \frac{2^{r+1}\ {}^{n}C_r}{r+1}$$

$$\frac{1}{n+1}\sum 2^{r+1}.\frac{n+1}{r+1}.\ {}^{n}C_r$$

$$\frac{1}{n+1}\sum_{r=0}^{n} 2^{r+1}\ {}^{n+1}C_{r+1}$$

$$\frac{1}{n+1}\left\{2.{}^{n+1}C_1 + 2^2\ {}^{n+1}C_2 + ... + 2^{n+1}\ {}^{n+1}C_{n+1}\right\}$$

$$+ {}^{n+1}C_0 \qquad\qquad - {}^{n+1}C_0$$

$$\frac{1}{n+1}\left\{(1+2)^{n+1}-1\right\}$$

$$\frac{1}{n+1}\left\{3^{n+1}-1\right\}$$

(d) $^{n}C_0 - \frac{^{n}C_1}{2} + \frac{^{n}C_2}{3} - .... + (-1)^n \frac{^{n}C_n}{n+1}$

$\sum_{r=0}^{n} (-1)^r \frac{^{n}C_r}{r+1}$

$\frac{1}{n+1} \sum_{r=0}^{n} (-1)^r \ {}^{n+1}C_{r+1}$

$\frac{1}{n+1} \left\{ {}^{n+1}C_1 - {}^{n+1}C_2 + {}^{n+1}C_3 - ..... (-1)^n \ {}^{n+1}C_{n+1} \right\}$

$\frac{1}{n+1} \{1\}$

6. Let $\sum_{r=0}^{100} \sum_{s=0}^{100} (C_r^2 + C_s^2 + C_r C_s) = m \left({}^{2n}C_n \right) + 2^p$ where m, n and p are even natural numbers and C_r represents the coefficient of x^r in the expansion of $(1 + x)^{100}$. Find the value of $(m + n + p)$.

माना $\sum_{r=0}^{100} \sum_{s=0}^{100} (C_r^2 + C_s^2 + C_r C_s) = m \left({}^{2n}C_n \right) + 2^p$ जहां m, n और p सम प्राकृत संख्याऐं हैं तथा $C_r (1+x)^{100}$ के प्रसार में x^r के गुणांक है। तब $(m + n + p)$ का मान ज्ञात कीजिए।

Ans. 502

Sol. $\sum_{r=0}^{100} \sum_{s=0}^{100} (C_r^2 + C_s^2 + C_r C_s) = \sum_{r=0}^{100} \left(\sum_{s=0}^{100} \left[\left({}^{n}C_r \right)^2 + \left({}^{n}C_s \right)^2 + {}^{n}C_r . {}^{n}C_s \right] \right)$

$$= \sum_{r=0}^{100} \left(101. \left({}^{n}C_r \right)^2 + \left[\underbrace{\left({}^{n}C_0 \right)^2 + \left({}^{n}C_1 \right)^2 + ... + \left({}^{n}C_{100} \right)^2}_{p} \right] + {}^{n}C_r \left[\underbrace{\left({}^{n}C_0 \right) + \left({}^{n}C_1 \right) + ... + \left({}^{n}C_{100} \right)}_{q} \right] \right) .....(A)$$

Now, for P : -

$\because (1+n)^n = {}^{n}C_0 + {}^{n}C_1 . x + ... {}^{n}C_n . x^n$ .....(1)

$(x+1)^n = {}^{n}C_0 . x^n + {}^{n}C_1 . x^{n-1} + ... + {}^{n}C_n$ .....(2)

Multiply equation (1) & (2) and compare coefficient of x^n both side, we get .....

$${}^{2n}C_n = \left({}^{n}C_0 \right)^2 + \left({}^{n}C_1 \right)^2 + ... C_0 = p$$

$p = {}^{200}C_{100}$

Put x = 1 in equation (1) we get

$2^n = 2^{100} = {}^nC_0 + {}^nC_1 + ...+ {}^nC_n = q$

Now, from (A)

$$\sum_{r=0}^{100}\left[101\times\left({}^nC_r\right)^2 + {}^{200}C_{100} + 2^{100}.\ {}^nC_r\right]$$

$$= 101\sum_{r=0}^{100}\left({}^nC_r\right)^2 + {}^{200}C_{100}\sum_{r=0}^{100}1 + 2^{100}\sum_{r=0}^{100}{}^nC_r$$

$= 101 \times {}^{200}C_{100} + {}^{200}C_{100} \times 101 + 2^{100} \times 2^{100}$

$= 202 \times {}^{200}C_{100} + 2^{200} = m({}^{2n}C_n) + 2^p$ (Given)

by comapre both side we get

m = 202, n = 100, p = 200

$\therefore$ m + n + p = 502

7. If $(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_n x^n$, then show that the sum of the products of the C_i's taken two at a time, represented by $\underset{0 \le i < j \le n}{\Sigma\,\Sigma\, C_i\, C_j}$ is equal to $2^{2n-1} - \frac{2n!}{2(n!)^2}$.

यदि $(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_n x^n$, तब सिद्ध कीजिए कि दो C_i's को एक साथ लेकर गुणन फलों का योग यदि $\underset{0 \le i < j \le n}{\Sigma\,\Sigma\, C_i\, C_j}$ से परिभाषित है, का मान $2^{2n-1} - \frac{2n!}{2(n!)^2}$ होगा।

Ans.

Sol. $(1+x)^n = C_0 + C_1x + ... + C_n x^n$

$S = \Sigma\Sigma\, C_iC_j = \frac{1}{2}\left[(C_0 + C_1 + C_2 + ...+ C_n)^2 - (C_0^2 + C_1^2 + C_2^2 + ...+ C_n^2)\right]$

$\Rightarrow 2S = 2^{2n} - {}^{2n}C_n \Rightarrow S = 2^{2n-1} - \frac{{}^{2n}C_n}{2}$

8. Let $S_1 = \sum\sum_{0\le i<j\le 100} C_i\, C_j$, $S_2 = \sum\sum_{0\le j<i\le 100} C_i\, C_j$ and $S_3 = \sum\sum_{0\le i=j\le 100} C_i\, C_j$

where C_r represents coefficient of x^r in the binomial expansion of $(1 + x)^{100}$.

If $S_1 + S_2 + S_3 = a^b$ where $a, b \in N$ then find the least value of $(a + b)$.

माना $S_1 = \sum\sum_{0\le i<j\le 100} C_i\, C_j$, $S_2 = \sum\sum_{0\le j<i\le 100} C_i\, C_j$ तथा $S_3 = \sum\sum_{0\le i=j\le 100} C_i\, C_j$

जहां C_r, $(1 + x)^{100}$ में x^r के गुणांक हैं, यदि $S_1 + S_2 + S_3 = a^b$ जहां $a, b \in N$ तब $(a + b)$ का न्यूनतम मान ज्ञात कीजिए।

Ans. 66

Sol. Given that $S_1 + S_2 + S_3 = \sum_{i=0}^{100}\sum_{j=0}^{100} C_i C_j$

$=\left({}^{100}C_0 + {}^{100}C_1 + {}^{100}C_2 + ... + {}^{100}C_{100}\right)^2$

$=\left(2^{100}\right)^2 = 2^{200} = a^b$

$\Rightarrow a^b = 2^{200} = 4^{100} = 16^{50} = 32^{40} = ...$

So, least value of a + b = 16 + 50 = 66

9. The expressions $1 + x$, $1+x + x^2$, $1 + x + x^2 + x^3$,..... $1 + x + x^2 +..... + x^n$ are multiplied together and the terms of the product thus obtained are arranged in increasing powers of x in the form of $a_0 + a_1x + a_2x^2 +....., then,

(a) how many terms are there in the product.

(b) show that the coefficients of the terms in the product, equidistant from the beginning and end are equal.

(c) show that the sum of the odd coefficients = the sum of the even coefficients = $\frac{(n+1)!}{2}$

यदि व्यंजक $1 + x$, $1+x + x^2$, $1 + x + x^2 + x^3$,..... $1 + x + x^2 +..... + x^n$ को गुणा किया जाता है तथा सभी पदों को x की बढ़ती घातों में निम्न $a_0 + a_1x + a_2x^2 +....., प्रकार से लिखा जाता है तो –

(a) गुणनफल में पदों की संख्या कितनी होगी।

(b) सिद्ध कीजिए कि गुणनफल में प्रारम्भ तथा अन्तः से समान दूरी पर स्थित पद समान है।

(c) सिद्ध कीजिए कि विषम गुणांकों का योग = सम गुणांकों का योग = $\frac{(n+1)!}{2}$

Ans. (a) $\frac{n^2+n+2}{2}$

Sol. $(1 + x)(1 + x + x^2)(1 + x + x^2 + x^3) ..... (1 + x + x^2 + ... + x^n)$

$= a_0 + a_1x + a_2x^2 + ... +$

(a) As the largest power of 'x' present in the expansion will be

$1 + 2 + 3 + ... + n$

$\Rightarrow \frac{n(n+1)}{2}$

So number of terms will be $1 + \frac{n(n+1)}{2}$

$\Rightarrow \frac{n^2+n+2}{2}$

(b) $(1 + x)\,(1 + x + x^2)\,(1 + x + x^2 + x^3)\,\ldots\ldots\,(1 + x + x^2 + \ldots\ldots + x^n)$

$= a_0 + a_1x + a_2x^2 + \ldots. + a_nx^{\frac{n(n+1)}{2}}$ ...(A)

Replace $x \to \frac{1}{x}$

$$\left(1+\frac{1}{x}\right)\left(1+\frac{1}{x}+\frac{1}{x^2}\right)\left(1+\frac{1}{x}+\frac{1}{x^2}+\frac{1}{x^3}\right).....\left(1+\frac{1}{x}+\frac{1}{x^2}+....+\frac{1}{x^n}\right)$$

$$= a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + ...\frac{a_n}{x^{\frac{n(n+1)}{2}}}$$

$$\frac{(x+1)}{x}\left(\frac{x^2+x+1}{x^2}\right)\left(\frac{x^3+x^2+x+1}{x^3}\right).....\left(\frac{x^n+x^{n-1}+....+1}{x^n}\right)$$

$$= \frac{a_0x^{\frac{n(n+1)}{2}} + a_1x^{\frac{n(n+1)}{2}-1} + ... + a_n}{x^{\frac{n(n+1)}{2}}}$$

$(1 + x)\,(1 + x + x^2)\,(1 + x + x^2 + x^3)\,\ldots\ldots\,(1 + x + x^2 + \ldots. + x^n) = a_0x^{\frac{n(n+1)}{2}} + a_1x^{\frac{n^2+n-2}{2}} + ... + a_n$ ...(B)

from (A) & (B)

$a_0 + a_1x + a_2x^2 + \ldots. + a_nx^{\frac{n(n+1)}{2}} = a_0x^{\frac{n(n+1)}{2}} + a_1x^{\frac{n^2+n-2}{2}} + \ldots + a_n$

Comparing coefficients

$$\left.\begin{array}{l} a_0 = a_n \\ a_1 = a_{n-1} \\ \vdots \\ a_n = a_0 \end{array}\right\} \Rightarrow a_r = a_{n-r}$$

(c) Put x = 1 in (A)

$2 \cdot 3 \cdot 4 \cdot \ldots\ldots\ldots (n + 1) = a_0 + a_1 + a_2 + \ldots\ldots\ldots + a_n$

Put $x = -1$ in (A)

$0 = a_0 - a_1 + a_2 - \ldots\ldots\ldots + a_n$

$$a_0 + a_2 + \ldots + a_n = \frac{(n+1)!}{2}$$

10. Let $x = \left(3\sqrt{6}+7\right)^{89}$. If $\{x\}$ denotes the fractional part of 'x' then find the remainder when $x\{x\} + (x\{x\})^2 + (x\{x\})^3$ is divided by 31.

माना $x = \left(3\sqrt{6}+7\right)^{89}$ है। यदि $\{x\}$, 'x' के भिन्नात्मक भाग को निरूपित करता है तो शेषफल ज्ञात कीजिए जब $x\{x\} + (x\{x\})^2 + (x\{x\})^3$, 31 से विभाजित किया जाये।

Ans. 0

Sol. $x = \left(3\sqrt{6}+7\right)^{89}$

Find $x\{x\} + \left(x\{x\}\right)^2 + \left(x\{x\}\right)^3$

$1 + y + y^2 + y^3 - 1$

$(1 + y) + y^2 (1 + y) - 1$

$(1 + y) (1 + y^2) - 1$

$$I + f = \left(7+3\sqrt{6}\right)^{89} = {}^{89}C_0 7^{89} + {}^{89}C_1 7^{88}\left(3\sqrt{6}\right) + \ldots\ldots$$

$$F = \left(7-3\sqrt{6}\right)^{89} = \ldots\ldots\ldots\ldots$$

$$2\left[{}^{89}C_0 7^{89} + {}^{89}C_2 7^{87}\left(3\sqrt{6}\right)^1 + \ldots\ldots {}^{89}C_{88}(7)^1\left(3\sqrt{6}\right)^{88}\right]$$

$I + f + F =$ Even Integer

$0 < f < 1$

$-1 < F < 0$

$-1 < f + F < 1$

$I =$ Even Integer

$\Rightarrow f + F = 0$

$\{x\} = -F$

$= -\left(7 - 3\sqrt{6}\right)^{89}$

$\{x\} = \left(3\sqrt{6} - 7\right)^{89}$

$x = \left(3\sqrt{6} + 7\right)^{89}$

$x\{x\} = \left\{\left(3\sqrt{6} - 7\right)\left(3\sqrt{6} + 7\right)\right\}^{89}$

$= \{54 - 49\}^{89}$

$= 5^{89}$

Now 5^{89} {5 + 25 + 125}

$5^{89} \times 5 \times 31$

Divide by 31

So remainder = 0

11. Find the sum of possible real values of x for which the sixth term of $\left(3^{\log_3\sqrt{9^{|x-2|}}} + 7^{\left(\frac{1}{5}\right)\log_7\left(3^{|x-2|}-9\right)}\right)^7$ equals 567.

x के संभव वास्तविक मानों का योग ज्ञात कीजिए, जिनके लिए $\left(3^{\log_3\sqrt{9^{|x-2|}}} + 7^{\left(\frac{1}{5}\right)\log_7\left(3^{|x-2|}-9\right)}\right)^7$ का छठा पद 567 के बराबर है।

Ans. 4

Sol. $\left\{3^{\log_3\left(9^{|x-2|}\right)^{\frac{1}{2}}} + 7^{\frac{1}{5}\log_7 3^{|x-2|}-9}\right\}^7$

$\left\{\left(9^{|x-2|}\right)^{\frac{1}{2}} + \left\{3^{|x-2|}-9\right\}^{\frac{1}{5}}\right\}^7$

$\left\{3^{|x-2|} + 3^{\frac{|x-2|-9}{5}}\right\}^7$

$$^{7}C_{5}\left(3^{|x-2|}\right)^{2}\left(3^{\frac{|x-2|-9}{5}}\right)^{5}=567$$

$$3^{2|x-2|}\cdot 3^{|x-2|-9}=\frac{567}{21}$$

$3^{|x-2|}=t$

$t^{2}\cdot\frac{t}{3^{9}}=27$

$t^{3}=3^{12}$

$t=3^{9}$

$3^{|x-2|}=3^{9}$

$|x-2|=9$

$x-2=\pm 9$

$x=2\pm 9$

$x=11,-7$

$11+(-7)$

$\Rightarrow 4$

12. Let $1+\sum_{r=1}^{10}\left(3^{r}.^{10}C_{r}+r.^{10}C_{r}\right)=2^{10}(\alpha\,.\,4^{5}+\beta)$ where $\alpha, \beta \in N$ and $f(x)=x^{2}-2x-k^{2}+1$. If α, β lies between the roots of $f(x)=0$. Then find the smallest positive integral value of k.

माना $1+\sum_{r=1}^{10}\left(3^{r}.^{10}C_{r}+r.^{10}C_{r}\right)=2^{10}(\alpha\,.\,4^{5}+\beta)$ है, जहाँ $\alpha, \beta \in N$ तथा $f(x)=x^{2}-2x-k^{2}+1$ है। यदि α, β, $f(x)=0$ के मूलों के बीच स्थित है, तो k का न्यूनतम संभव पूर्णांक मान ज्ञात कीजिए।

Ans. 5

Sol. $1+\sum_{r=1}^{10}3^{r}\ {}^{10}C_{r}+\sum_{r=1}^{10}r\ {}^{10}C_{r}$

$$=1+3^{0}\ {}^{10}C_{0}+3^{1}\ {}_{10}C_{1}+3^{2}\ {}^{10}C_{2}+....+3^{10}\ {}^{10}C_{10}-3^{0}.^{10}C_{0}+\sum_{r=1}^{10}r.^{10}C_{r}$$

$$=(1+3)^{10}+\sum_{r=1}^{10}r\cdot\frac{10}{r}\cdot^{9}C_{r-1}$$

$$=4^{10}+10\left[^{9}C_{0}+^{9}C_{1}+^{9}C_{2}+.....+^{9}C_{9}\right]$$

$$=4^{10}+10\left(2^{9}\right)$$

$= 4^{10} + 10 \cdot 2^9$

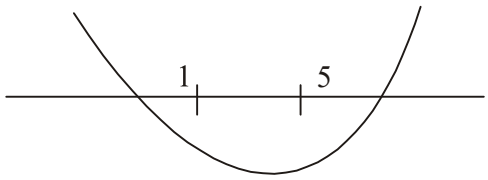
$= 4^{10} + 5 \cdot 2^{10}$

$\Rightarrow 2^{10}\left[4^5 + 5\right] = 2^{10}\left[\alpha \cdot 4^5 + \beta\right]$

$\alpha = 1, \beta = 5$

Now,

$f(x) = x^2 - 2x - k^2 + 1$



$f(1) < 0$ $\qquad$ $f(5) < 0$

$1 - 2 - k^2 + 1 < 0$ $\qquad$ $25 - 10 - k^2 + 1 < 0$

$-k^2 < 0$ $\qquad$ $16 - k^2 < 0$

$k^2 > 0$ $\qquad$ $k \in (-\infty, -4) \cup (4, \infty)$

$k \neq 0$

Smallest +ve integral value = 5

13. Let $n \in N$; $S_n = \sum_{r=0}^{3n}\left({}^{3n}C_r\right)$ and $T_n = \sum_{r=0}^{n}\left({}^{3n}C_{3r}\right)$. Find $|S_n - 3T_n|$.

माना $n \in N$; $S_n = \sum_{r=0}^{3n}\left({}^{3n}C_r\right)$ तथा $T_n = \sum_{r=0}^{n}\left({}^{3n}C_{3r}\right)$ है, तो $|S_n - 3T_n|$ का मान ज्ञात कीजिए।

Ans. 2

Sol. $S_n = \sum_{r=0}^{3n} {}^{3n}C_r$

$= {}^{3n}C_0 + {}^{3n}C_1 + {}^{3n}C_2 + ... + {}^{3n}C_{3n}$ ...(A)

$(1 + x)^{3n} = {}^{3n}C_0\, x^0 + {}^{3n}C_1\, x^1 + ... + {}^{3n}C_{3n}\, x^{3n}$

Put $x = 1$

$S_n = 2^{3n}$

$T_n = \sum_{r=0}^{n} {}^{3n}C_{3r}$

$= {}^{3n}C_0 + {}^{3n}C_3 + {}^{3n}C_6 + ... + {}^{3n}C_{3n}$

$(1 + x)^{3n} = {}^{3n}C_0\, x^0 + {}^{3n}C_1\, x + {}^{3n}C_2\, x^2 + {}^{3n}C_3\, x^3 + {}^{3n}C_4\, x^4 + ... + {}^{3n}C_{3n}\, x^{3n}$

$x = w$

$(1 + w)^{3n} = {}^{3n}C_0 + {}^{3n}C_1\, w + {}^{3n}C_2\, w^2 + {}^{3n}C_3 + ... + {}^{3n}C_{3n}$

$x = w^2$

$(1 + w^2)^{3n} = {}^{3n}C_0 + {}^{3n}C_1\, w^2 + {}^{3n}C_2\, w + {}^{3n}C_3 + ... + {}^{3n}C_{3n}$

$(1 + w)^{3n} + (1 + w^2)^{3n} = 2\ {}^{3n}C_0 + {}^{3n}C_1\,(-1) + {}^{3n}C_2\,(-1) + 2\ {}^{3n}C_3 + ...$

Now Add (A) in it

$2^{3n} + (1 + w)^{3n} + (1 + w^2)^{3n} = 3[T_n]$

$2^{3n} + (-w^2)^{3n} + (-w)^{3n} = 3T_n$

$2^{3n} + (-1)^{3n} + (-1)^{3n} = 3T_n$

$|S_n - 3T_n|$

$= |(-1)^{3n} + (-1)^{3n}|$

$\Rightarrow 2$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 14 to 16

Consider the binomial expansion $R = (1 + 2x)^n = I + f$ where I is the integral part of R and f is the fractional part of R $n \in N$. Also the sum of the coefficients of R is 6561.

माना द्विपद प्रसार $R = (1 + 2x)^n = I + f$ है, जहाँ I तथा f, क्रमशः R के पूर्णांक भाग तथा भिन्नात्मक भाग है $n \in N$. R के गुणांकों का योग 6561 है।

14. The value of $(n + R - Rf)$ for $x = \frac{1}{\sqrt{2}}$ equals :

$x = \frac{1}{\sqrt{2}}$ के लिए $(n + R - Rf)$ का मान होगा:

(A) 7 (B) 8 (C) 9 (D) 10

Ans. C

Sol. $I + f = (1 + 2x)^n = R$

Sum of coefficient 6561

Put $x = 1$

$3^n = 6561 = 3^8$

$n = 8$

$R = I + f = \left(1+\sqrt{2}\right)^8$

$F = \left(1-\sqrt{2}\right)^8$

$I + f = {}^8C_0 + {}^8C_1\sqrt{2} + ... + {}^8C_8\left(\sqrt{2}\right)^8$

$F = {}^8C_0 - {}^8C_1\sqrt{2} ..... + {}^8C_8\left(\sqrt{2}\right)^8$

$I + f + F = 2[{}^8C_0 + {}^8C_2 \cdot 2 + {}^8C_4\, 2^2 + {}^8C_6\, 2^3 + {}^8C_8\, 2^4]$

$I + f + F = 2 \times 577 = 1154$

$0 < f < 1$

$0 < F < 1$

$0 < f + F < 2$

$f + F = 1$

$I = 1153$

Now $n + R(1 - f)$

$8 + \left(1+\sqrt{2}\right)^8\left(1-\sqrt{2}\right)^8$

$8 + \{1 - 2\}^8$

9

15. If i^{th} terms is the greatest term for $x = 1/2$, then 'i' equals :

यदि $x = 1/2$ के लिए iवाँ पद महतम पद है, तो 'i' बराबर होगा :

(A) 4 (B) 5 (C) 6 (D) 7

Ans. B

Sol. $\frac{n+1}{1+\left|\frac{a}{b}\right|} - 1 \le i \le \frac{n+1}{1+\left|\frac{a}{b}\right|}$

Now $\frac{9}{1+\left|\frac{1}{2x}\right|} \Rightarrow \frac{9}{1+1} \Rightarrow \frac{9}{2} \Rightarrow 4.5$

$3.5 \le i \le 4.5$

$\Rightarrow i = 4$

So term will be 5th term.

16. If k^{th} terms is having greatest coefficient then sum of all possible value(s) of k is :

यदि kवें पद का गुणांक महतम है, तो k के सभी संभव मानों का योग होगा :

(A) 6 (B) 7 (C) 11 (D) 13

Ans. D

Sol. $n = 8$

So, greatest coefficient will be middle term.

Paragraph for Question Nos. 17 to 19

Let us consider the expansion $(3 + 4x)^n = a_0 + a_1x + a_2x^2 + a_3x^3 + ... + a_nx^n$, then

माना प्रसार $(3 + 4x)^n = a_0 + a_1x + a_2x^2 + a_3x^3 + ... + a_nx^n$ है, तो

17. $a_1 + a_3 + a_5 + a_7 + ...$ is equal to :

$a_1 + a_3 + a_5 + a_7 + ...$ बराबर है :

(A) $\frac{7^n + (-1)^n}{2}$ (B) $\frac{7^n - (-1)^n}{2}$ (C) $\frac{7^{n-1} - (-1)^{n-1}}{2}$ (D) $\frac{7^{n-1} + (-1)^{n-1}}{2}$

Ans. B

Sol. $(3 + 4x)^n = a_0 + a_1x + a_2x^2 + ... + a_nx^n$

$x = 1$

$7^n = a_0 + a_1 + a_2 + a_3 + ... + a_n$

$x = -1$

$(-1)^n = a_0 - a_1 + a_2 - a_3 - .....$

$$\frac{7^n + (-1)^n}{2} = a_0 + a_2 + a_4 + ...$$

$$\frac{7^n - (-1)^n}{2} = a_1 + a_3 + a_5 + ...$$

18. $a_0 - a_2 + a_4 - a_6 + ...$ is equal to :

$a_0 - a_2 + a_4 - a_6 + ...$ बराबर है :

(A) $5^n \cos\left(n \tan^{-1}\frac{4}{3}\right)$ (B) $7^n \cos\left(n \tan^{-1}\frac{3}{4}\right)$ (C) $7^n \cos\left(n \tan^{-1}\frac{4}{3}\right)$ (D) $5^n \cos\left(n \tan^{-1}\frac{3}{4}\right)$

Ans. A

Sol. $(3 + 4i)^n = a_0 + a_1 i - a_2 - a_3 i + a_4 + a_5 i - a_6 .....$

$(3 + 4i)^n = (a_0 - a_2 + a_4 - a_6 + ...) + i(a_1 - a_3 + a_5 - a_7 + ...)$

$$\left\{5e^{i\left(\tan^{-1}\frac{4}{3}\right)}\right\}^n$$

$$5^n e^{in \tan^{-1}\frac{4}{3}}$$

$$5^n \left\{\cos\left(n \tan^{-1}\frac{4}{3}\right) + i \sin\left(n \tan^{-1}\frac{4}{3}\right)\right\}$$

$= (a_0 - a_2 + a_4 - a_6 + ...) + i(a_1 - a_3 + a_5 - a_7 + ...)$

Now $a_0 - a_2 + a_4 - a_6 + ... = 5^n \cos\left(n \tan^{-1}\frac{4}{3}\right)$

19. $(a_0 - a_2 + a_4 - a_6 + ...)^2 + (a_1 - a_3 + a_5 - .....)^2$ equals :

$(a_0 - a_2 + a_4 - a_6 + ...)^2 + (a_1 - a_3 + a_5 - .....)^2$ बराबर है :

(A) 7^{2n} (B) 5^n (C) 5^{2n} (D) 7^n

Ans. C

Sol. $(a_0 - a_2 + a_4 - a_6 + ...)^2 + (a_1 - a_3 + a_5 - a_7 + ...)$

$= 5^{2n}$ [from above $\sin^2\theta + \cos^2\theta = 1$]

MATCH THE COLUMN TYPE QUESTION(S)

20. If $\left(1 + x + x^2\right)^n = a_0 + a_1 x + a_2 x^2 + ... + a_{2n} x^{2n}$, then :

यदि $\left(1 + x + x^2\right)^n = a_0 + a_1 x + a_2 x^2 + ... + a_{2n} x^{2n}$, तो:

Column -I	**Column-II**
(A) $a_0 + a_1 + ...a_{2n}$	(P) n+1
(B) $a_0 - a_1 + a_2 - a_3 + ... + a_{2n}$	(Q) 1

(C) $a_0 - 2a_1 + 3a_2 + \ldots\ldots + (2n+1)a_{2n}$ (R) $3n$

(D) $a_0 - 2a_1 + 4a_2 - 8a_3 + \ldots\ldots + 2^{2n}.a_{2n}$ (S) 3^n

Ans. A – S, B – Q, C – P, D – S

Sol. $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + \ldots + a_{2n}x^{2n}$

$[A] \rightarrow S$

$x = 1$

$3^n = a_0 + a_1 + a_2 + \ldots\ldots + a_{2n}$

$[B] \rightarrow Q$

$x = -1$

$(1 - 1 + 1)^n = a_0 - a_1 + a_2 - a_3 + \ldots\ldots + a_{2n}$

$= 1$

$[C] \rightarrow Q$

$(1 + x + x^2)^n \cdot x = a_0x + a_1x^2 + a_2x^3 + \ldots\ldots + a_2nx^{2n+1} = (x^n + x^{n+1} + x^{n+2})^n$

Now differentiating

$n(x^n + x^{n+1} + x^{n+2})^{n-1} \{nx^{n-1} + (n+1)x^n + (n+2)x^{n+1}\} = a_0 + 2a_1x + 3a_2x^2 + \ldots + (2n+1)a_{2n}x^{2n}$

Put $x = -1$

$n((-1)^n + (-1)^{n+1} + (-1)^n)^{n-1} \{n(-1)^{n-1} + (n+1)(-1)^n + (n+2)(-1)^{n+1}\}$

$= a_0 - 2a_1 + 3a_2 - \ldots\ldots\ldots\ldots + (2n+1)a_{2n}$

$n(-1 + 1 - 1)^{even} \{+n - n - 1 + n + 2\}$

$n(n+1)$